

Cyber Security Task 2 (2026)

Email Phishing Analysis & Security Awareness Report

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INTRODUCTION

Phishing is a social engineering attack where cybercriminals send fake emails or messages to trick users into revealing confidential information such as passwords, banking details, and login credentials.

These attacks often use fake websites, malicious attachments, and urgent messages to manipulate victims. Phishing mainly targets human mistakes rather than technical vulnerabilities.

This project focuses on identifying phishing characteristics, analyzing suspicious emails, classifying email threats, and spreading awareness about safe email practices.

OBJECTIVE

Objective:

To examine phishing and suspicious email samples, detect common phishing indicators, analyze risk levels, and improve user awareness regarding email security threats.

TOOLS USED

- * Microsoft Word
- * Web Browser
- * Google Header Analyzer
- * MXToolbox
- * Screenshot Utility

EMAIL SAMPLE 1 — PHISHING EMAIL

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File Edit View H1 H2 H3 H4 H5 H6 H7 H8 H9 H10 B I S G L T
Urgent: Account Access Verification Required
Dear User,

We noticed multiple failed login attempts on your online banking account from an unknown device.

For your protection, access to your account has been limited temporarily.

Please confirm your identity and restore account access by visiting the secure verification page below:
http://account-secure-check[.]net

If verification is not completed within 12 hours, your online banking services may be disabled.

Thank you,
Customer Protection Department
```

Analysis

Email Classification: PHISHING

Risk Level: HIGH

Phishing Indicators

1. Generic Greeting

- * Uses "Dear User" instead of actual customer name

2. Fear & Urgency

- * Threatens account deactivation

3. Suspicious Domain

- * Fake banking verification website

4. Non-Secure Link

- * Uses HTTP connection

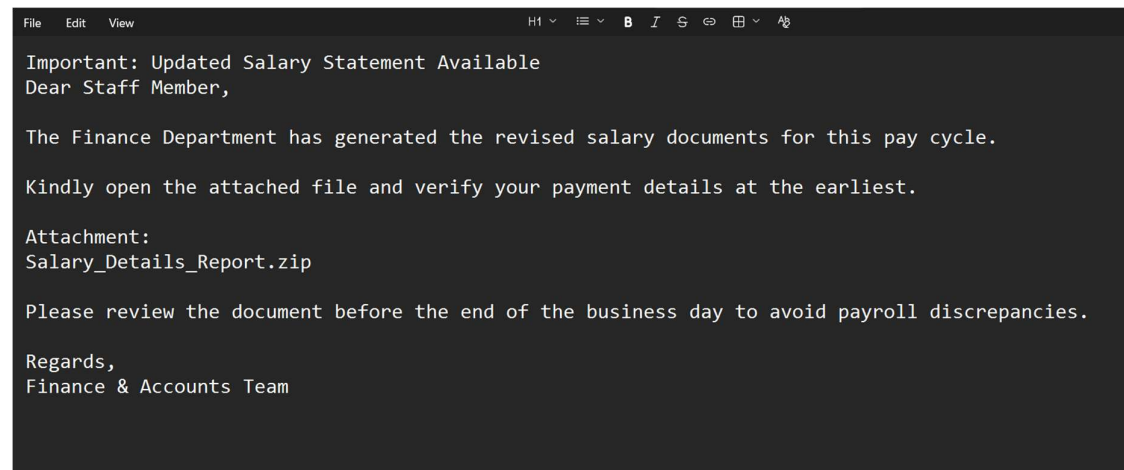
Explanation

This email attempts to create panic and pressure the recipient into clicking a malicious banking link. Attackers commonly use urgent language to steal login credentials and financial information.

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EMAIL SAMPLE 2 — SUSPICIOUS EMAIL

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Analysis

Email Classification: SUSPICIOUS
Risk Level: MEDIUM

Indicators

1. Compressed ZIP Attachment
 - * May contain harmful files
2. Generic Greeting
 - * Uses "Hello Staff Member"
3. Immediate Action Request
 - * Encourages quick download
4. Unexpected File
 - * Unusual attachment format

Explanation

This email includes a ZIP attachment that could potentially contain malware or malicious scripts. Users should always verify unexpected attachments before opening them.

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EMAIL SAMPLE 3 — SAFE EMAIL

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Notice: Microsoft Account Protection Settings Modified

Hello User,

We're writing to inform you that changes were recently made to your Microsoft account protection preferences on May 18, 2026.

If you authorized these updates, you do not need to take any additional action.

To review recent sign-in activity and security settings, visit the link below:

https://account.microsoft.com/security

If you do not recognize this activity, we recommend securing your account immediately.

Sincerely,
Microsoft Account Security Team
```

Analysis

Email Classification: SAFE
Risk Level: LOW

Indicators

1. Official Domain

- * Uses legitimate microsoft.com domain

2. Secure HTTPS URL

- * Encrypted and trusted connection

3. Professional Communication

- * No threatening language

4. Informational Tone

- * Does not pressure the user

Explanation

This email appears legitimate because it uses Microsoft's official domain, secure HTTPS connection, and a professional tone without suspicious urgency or threats.

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SECURITY AWARENESS GUIDELINES

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- * Avoid clicking unknown or suspicious links
- * Verify sender email addresses carefully
- * Never download unexpected attachments
- * Enable Multi-Factor Authentication (MFA)
- * Use strong passwords
- * Report suspicious emails immediately
- * Confirm website legitimacy before entering credentials

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EMPLOYEE BEST PRACTICES

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DO'S:

- * Verify official domains
- * Enable MFA
- * Scan attachments before opening
- * Report suspicious emails

DON'TS:

- * Don't trust urgent messages blindly
- * Don't share passwords or OTPs
- * Don't click unknown links
- * Don't download unverified files

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CONCLUSION

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This project explained how phishing attacks manipulate users through fake websites, suspicious attachments, and urgency tactics.

By understanding phishing indicators and following safe email practices, users can reduce cybersecurity risks and better protect personal and organizational information.